

Linear Programming Course Summary

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1 Key Terms

1.1 linear programming fundamentals

1.2 solution space analysis

- **Feasible Region**: *The set of all points that satisfy all the constraints of a linear programming problem.*

1.3 simplex method algorithms

2 Problem Types

2.1 linear programming fundamentals

2.2 advanced linear programming

- **Handling Infeasible Origins:** *The origin $(0,0)$ may not satisfy all constraints in a linear programming problem. Techniques like introducing an auxiliary problem are used to find a feasible starting point.*

3 Algorithm Solutions

3.1 linear optimization techniques

- **Auxiliary Problem:** *When the origin is not feasible, introduce a new variable x_0 and modify the constraints to create an auxiliary problem that maximizes $-x_0$. Solve this auxiliary problem using the Simplex method. If the optimal solution has $x_0 = 0$, then the solution is feasible for the original problem.*